SCR2043 OPERATING SYSTEMS

Name	:	Damiya Aina binti Basir Abd Shammad	Marks
Student ID	:	A23CS0220	
Section	:	02	

This lab assessment is designed to test your understanding and skills on some basic concepts and tools related to process monitoring and management in operating system. Please follow the instructions carefully and submit your answers in this word document and rename the file as **os-lab-assessment02-studentname-matricno.docx**.

Essential Steps Before Starting Lab Assessment 2:

1. **Download necessary source codes:**

   Use the `wget` command to retrieve the following source code files to your Linux (or WSL or MacOS) environment:

```
wget -O mainprocess.c https://rebrand.ly/mainprocess_c
wget -O subprocess1.c https://rebrand.ly/subprocess1_c
wget -O subprocess2.c https://rebrand.ly/subprocess2_c
```

2. **Compile the source files:**

   Use the `gcc` compiler to create executable files from the source code.

```
gcc mainprocess.c -o mainprocess
gcc subprocess1.c -o subprocess1
gcc subprocess2.c -o subprocess2
```

3. **Execute the dummy processes:**

   Run all the dummy processes

```
./mainprocess &
```

   Press **enter** two times.

4. The dummy processes are running for 2 hours. If you took longer than 2 hours on questions 1-9, please restart the main process with `./mainprocess &`.

Lab Assessment 2 : Linux Process Monitoring and Management

Instructions:

1. Carefully execute each command as instructed in the questions.
2. Write down the exact command used for each task.
3. Capture a screenshot of the command's output.

Question 1

Use the `ps` command with the appropriate option to display a complete list of all running processes within the Linux operating system.

Command

```
aina@secr2043:~$ ps -e
```

Output

```
    PID TTY          TIME CMD
      1 ?        00:00:00 systemd
      2 ?        00:00:00 kthreadd
      3 ?        00:00:00 pool_workqueue_release
      4 ?        00:00:00 kworker/R-rcu_g
      5 ?        00:00:00 kworker/R-rcu_p
      6 ?        00:00:00 kworker/R-slub_
      7 ?        00:00:00 kworker/R-netns
      8 ?        00:00:00 kworker/0:0-cgroup_destroy
      9 ?        00:00:00 kworker/0:1-events
     10 ?        00:00:00 kworker/0:0H-kblockd
     11 ?        00:00:00 kworker/u4:0-floppy
     12 ?        00:00:00 kworker/R-mm_pe
     13 ?        00:00:00 rcu_tasks_kthread
     14 ?        00:00:00 rcu_tasks_rude_kthread
     15 ?        00:00:00 rcu_tasks_trace_kthread
     16 ?        00:00:00 ksoftirqd/0
     17 ?        00:00:00 rcu_preempt
     18 ?        00:00:00 migration/0
     19 ?        00:00:00 idle_inject/0
     20 ?        00:00:00 cpuhp/0
     21 ?        00:00:00 cpuhp/1
     22 ?        00:00:00 idle_inject/1
     23 ?        00:00:00 migration/1
     24 ?        00:00:00 ksoftirqd/1
     25 ?        00:00:00 kworker/1:0-events
     26 ?        00:00:00 kworker/1:0H-events_highpri
     27 ?        00:00:00 kworker/u5:0-events_unbound
     28 ?        00:00:00 kworker/u6:0-events_power_efficient
     29 ?        00:00:00 kdevtmpfs
```

```
     30 ?        00:00:00 kworker/R-inet_
     31 ?        00:00:00 kworker/u5:1-flush-8:0
     32 ?        00:00:00 kauditd
     33 ?        00:00:00 khungtaskd
     34 ?        00:00:00 oom_reaper
     35 ?        00:00:00 kworker/R-write
     36 ?        00:00:00 kworker/u5:2-events_unbound
     37 ?        00:00:00 kcompactd0
     38 ?        00:00:00 ksmd
     39 ?        00:00:00 kworker/1:1-events
     40 ?        00:00:00 khugepaged
     41 ?        00:00:00 kworker/R-kinte
     42 ?        00:00:00 kworker/R-kbloc
     43 ?        00:00:00 kworker/R-blkcg
     44 ?        00:00:00 irq/9-acpi
     45 ?        00:00:00 kworker/R-tpm_d
     46 ?        00:00:00 kworker/R-ata_s
     47 ?        00:00:00 kworker/R-md
     48 ?        00:00:00 kworker/R-md_bi
     49 ?        00:00:00 kworker/R-edac-
     50 ?        00:00:00 kworker/R-devfr
     51 ?        00:00:00 watchdogd
     52 ?        00:00:00 kworker/u6:1-events_unbound
     53 ?        00:00:00 kworker/1:1H-kblockd
     54 ?        00:00:00 kswapd0
     55 ?        00:00:00 ecryptfs-kthread
     56 ?        00:00:00 kworker/R-kthro
     57 ?        00:00:00 kworker/R-acpi_
     58 ?        00:00:00 kworker/u6:2-flush-8:0
     59 ?        00:00:00 scsi_eh_0
```

```
     60 ?        00:00:00 kworker/R-scsi_
     61 ?        00:00:00 scsi_eh_1
     62 ?        00:00:00 kworker/R-scsi_
     63 ?        00:00:00 kworker/u6:3-events_power_efficient
     64 ?        00:00:00 kworker/u6:4-events_power_efficient
     65 ?        00:00:00 kworker/R-mld
     66 ?        00:00:00 kworker/R-ipv6_
     73 ?        00:00:00 kworker/R-kstrp
     75 ?        00:00:00 kworker/u7:0
     76 ?        00:00:00 kworker/u8:0
     77 ?        00:00:00 kworker/u9:0
     82 ?        00:00:00 kworker/R-crypt
     83 ?        00:00:00 kworker/0:1H-kblockd
     84 ?        00:00:00 kworker/1:2
     93 ?        00:00:00 kworker/R-charg
    132 ?        00:00:00 kworker/u4:1-ext4-rsv-conversion
    133 ?        00:00:00 kworker/0:2-events
    143 ?        00:00:00 kworker/R-mpt_p
    149 ?        00:00:00 kworker/R-mpt/0
    152 ?        00:00:00 kworker/0:3-cgroup_destroy
    157 ?        00:00:00 kworker/0:2H
    158 ?        00:00:00 scsi_eh_2
    159 ?        00:00:00 kworker/R-scsi_
    193 ?        00:00:00 kworker/R-raid5
    230 ?        00:00:00 jbd2/sda1-8
    231 ?        00:00:00 kworker/R-ext4-
    298 ?        00:00:00 psimon
    303 ?        00:00:00 systemd-journal
    323 ?        00:00:00 kworker/u5:3-events_unbound
    333 ?        00:00:00 kworker/R-kmpat
```

```
    340 ?        00:00:00 kworker/R-kmpat
    341 ?        00:00:00 kworker/0:4-events
    342 ?        00:00:00 kworker/0:5-cgroup_destroy
    343 ?        00:00:00 kworker/0:6
    344 ?        00:00:00 kworker/0:7-events
    364 ?        00:00:00 multipathd
    372 ?        00:00:00 systemd-udevd
    384 ?        00:00:00 psimon
    459 ?        00:00:00 jbd2/sda16-8
    460 ?        00:00:00 kworker/R-ext4-
    491 ?        00:00:00 systemd-network
    498 ?        00:00:00 systemd-resolve
    500 ?        00:00:00 irq/18-vmwgfx
    501 ?        00:00:00 kworker/R-ttm
    513 ?        00:00:00 systemd-timesyn
    647 ?        00:00:00 dbus-daemon
    651 ?        00:00:00 polkitd
    657 ?        00:00:00 systemd-logind
    658 ?        00:00:00 udisksd
    695 ?        00:00:00 unattended-upgr
    697 ?        00:00:00 rsyslogd
    712 ?        00:00:00 ModemManager
    779 ?        00:00:00 cron
    787 tty1     00:00:00 agetty
    850 ?        00:00:00 sshd
    851 ?        00:00:00 sshd
    853 ?        00:00:00 sshd
    854 pts/0    00:00:00 bash
    871 ?        00:00:01 fwupd
    922 pts/0    00:00:00 mainprocess
```

```
    923 pts/0    00:00:00 mainprocess
    924 pts/0    00:00:00 subprocess1
    925 pts/0    00:00:00 mainprocess
    926 pts/0    00:00:00 subprocess2
    927 pts/0    00:00:00 subprocess2
    928 pts/0    00:00:00 subprocess2
    929 pts/0    00:00:00 subprocess1
    930 pts/0    00:00:00 ps
```

Question 2

Employ the `ps` command with necessary options to unveil comprehensive details about each running process.

Command
`aina@secr2043:~$ ps -ef`

Output

```
UID          PID    PPID  C STIME TTY          TIME CMD
root           1       0  0 14:01 ?        00:00:00 /sbin/init
root           2       0  0 14:01 ?        00:00:00 [kthreadd]
root           3       2  0 14:01 ?        00:00:00 [pool_workqueue_release]
root           4       2  0 14:01 ?        00:00:00 [kworker/R-rcu_g]
root           5       2  0 14:01 ?        00:00:00 [kworker/R-rcu_p]
root           6       2  0 14:01 ?        00:00:00 [kworker/R-slub_]
root           7       2  0 14:01 ?        00:00:00 [kworker/R-netns]
root           8       2  0 14:01 ?        00:00:00 [kworker/0:0-cgroup_destroy]
root           9       2  0 14:01 ?        00:00:00 [kworker/0:1-events]
root          10       2  0 14:01 ?        00:00:00 [kworker/0:0H-kblockd]
root          11       2  0 14:01 ?        00:00:00 [kworker/u4:0-floppy]
root          12       2  0 14:01 ?        00:00:00 [kworker/R-mm_pe]
root          13       2  0 14:01 ?        00:00:00 [rcu_tasks_kthread]
root          14       2  0 14:01 ?        00:00:00 [rcu_tasks_rude_kthread]
root          15       2  0 14:01 ?        00:00:00 [rcu_tasks_trace_kthread]
root          16       2  0 14:01 ?        00:00:00 [ksoftirqd/0]
root          17       2  0 14:01 ?        00:00:00 [rcu_preempt]
root          18       2  0 14:01 ?        00:00:00 [migration/0]
root          19       2  0 14:01 ?        00:00:00 [idle_inject/0]
root          20       2  0 14:01 ?        00:00:00 [cpuhp/0]
root          21       2  0 14:01 ?        00:00:00 [cpuhp/1]
root          22       2  0 14:01 ?        00:00:00 [idle_inject/1]
root          23       2  0 14:01 ?        00:00:00 [migration/1]
root          24       2  0 14:01 ?        00:00:00 [ksoftirqd/1]
root          25       2  0 14:01 ?        00:00:00 [kworker/1:0-cgroup_destroy]
root          26       2  0 14:01 ?        00:00:00 [kworker/1:0H-events_highpri]
root          28       2  0 14:01 ?        00:00:00 [kworker/u6:0-events_power_efficient]
root          29       2  0 14:01 ?        00:00:00 [kdevtmpfs]
root          30       2  0 14:01 ?        00:00:00 [kworker/R-inet_]
root          31       2  0 14:01 ?        00:00:00 [kworker/u5:1-writeback]
root          32       2  0 14:01 ?        00:00:00 [kauditd]
root          33       2  0 14:01 ?        00:00:00 [khungtaskd]
root          34       2  0 14:01 ?        00:00:00 [oom_reaper]
root          35       2  0 14:01 ?        00:00:00 [kworker/R-write]
root          37       2  0 14:01 ?        00:00:00 [kcompactd0]
root          38       2  0 14:01 ?        00:00:00 [ksmd]
root          39       2  0 14:01 ?        00:00:02 [kworker/1:1-events]
root          40       2  0 14:01 ?        00:00:00 [khugepaged]
root          41       2  0 14:01 ?        00:00:00 [kworker/R-kinte]
root          42       2  0 14:01 ?        00:00:00 [kworker/R-kbloc]
root          43       2  0 14:01 ?        00:00:00 [kworker/R-blkcg]
root          44       2  0 14:01 ?        00:00:00 [irq/9-acpi]
root          45       2  0 14:01 ?        00:00:00 [kworker/R-tpm_d]
root          46       2  0 14:01 ?        00:00:00 [kworker/R-ata_s]
root          47       2  0 14:01 ?        00:00:00 [kworker/R-md]
root          48       2  0 14:01 ?        00:00:00 [kworker/R-md_bi]
root          49       2  0 14:01 ?        00:00:00 [kworker/R-edac-]
root          50       2  0 14:01 ?        00:00:00 [kworker/R-devfr]
root          51       2  0 14:01 ?        00:00:00 [watchdogd]
root          52       2  0 14:01 ?        00:00:00 [kworker/u6:1-events_unbound]
root          53       2  0 14:01 ?        00:00:00 [kworker/1:1H-kblockd]
root          54       2  0 14:01 ?        00:00:00 [kswapd0]
root          55       2  0 14:01 ?        00:00:00 [ecryptfs-kthread]
root          56       2  0 14:01 ?        00:00:00 [kworker/R-kthro]
root          57       2  0 14:01 ?        00:00:00 [kworker/R-acpi_]
root          58       2  0 14:01 ?        00:00:00 [kworker/u6:2-events_power_efficient]
root          59       2  0 14:01 ?        00:00:00 [scsi_eh_0]
root          60       2  0 14:01 ?        00:00:00 [kworker/R-scsi_]
root          61       2  0 14:01 ?        00:00:00 [scsi_eh_1]
```

```
root          62       2  0 14:01 ?        00:00:00 [kworker/R-scsi_]
root          64       2  0 14:01 ?        00:00:00 [kworker/u6:4-events_unbound]
root          65       2  0 14:01 ?        00:00:00 [kworker/R-mld]
root          66       2  0 14:01 ?        00:00:00 [kworker/R-ipv6_]
root          73       2  0 14:01 ?        00:00:00 [kworker/R-kstrp]
root          75       2  0 14:01 ?        00:00:00 [kworker/u7:0]
root          76       2  0 14:01 ?        00:00:00 [kworker/u8:0]
root          77       2  0 14:01 ?        00:00:00 [kworker/u9:0]
root          82       2  0 14:01 ?        00:00:00 [kworker/R-crypt]
root          83       2  0 14:01 ?        00:00:00 [kworker/0:1H-kblockd]
root          93       2  0 14:01 ?        00:00:00 [kworker/R-charg]
root         132       2  0 14:01 ?        00:00:00 [kworker/u4:1-ext4-rsv-conversion]
root         143       2  0 14:01 ?        00:00:00 [kworker/R-mpt_p]
root         149       2  0 14:01 ?        00:00:00 [kworker/R-mpt/0]
root         158       2  0 14:01 ?        00:00:00 [scsi_eh_2]
root         159       2  0 14:01 ?        00:00:00 [kworker/R-scsi_]
root         193       2  0 14:01 ?        00:00:00 [kworker/R-raid5]
root         230       2  0 14:01 ?        00:00:00 [jbd2/sda1-8]
root         231       2  0 14:01 ?        00:00:00 [kworker/R-ext4-]
root         298       2  0 14:01 ?        00:00:00 [psimon]
root         303       1  0 14:01 ?        00:00:00 /usr/lib/systemd/systemd-journald
root         323       2  0 14:01 ?        00:00:00 [kworker/u5:3-events_power_efficient]
root         333       2  0 14:01 ?        00:00:00 [kworker/R-kmpat]
root         340       2  0 14:01 ?        00:00:00 [kworker/R-kmpat]
root         364       1  0 14:01 ?        00:00:00 /sbin/multipathd -d -s
root         372       1  0 14:01 ?        00:00:00 /usr/lib/systemd/systemd-udevd
root         384       2  0 14:01 ?        00:00:00 [psimon]
root         459       2  0 14:01 ?        00:00:00 [jbd2/sda16-8]
root         460       2  0 14:01 ?        00:00:00 [kworker/R-ext4-]
systemd+     491       1  0 14:01 ?        00:00:00 /usr/lib/systemd/systemd-networkd
systemd+     498       1  0 14:01 ?        00:00:00 /usr/lib/systemd/systemd-resolved
root         500       2  0 14:01 ?        00:00:00 [irq/18-vmwgfx]
root         501       2  0 14:01 ?        00:00:00 [kworker/R-ttm]
systemd+     513       1  0 14:01 ?        00:00:00 /usr/lib/systemd/systemd-timesyncd
message+     647       1  0 14:01 ?        00:00:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --syst
polkitd      651       1  0 14:01 ?        00:00:00 /usr/lib/polkit-1/polkitd --no-debug
root         657       1  0 14:01 ?        00:00:00 /usr/lib/systemd/systemd-logind
root         658       1  0 14:01 ?        00:00:00 /usr/libexec/udisks2/udisksd
root         695       1  0 14:01 ?        00:00:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-s
syslog       697       1  0 14:01 ?        00:00:00 /usr/sbin/rsyslogd -n -iNONE
root         712       1  0 14:01 ?        00:00:00 /usr/sbin/ModemManager
root         779       1  0 14:01 ?        00:00:00 /usr/sbin/cron -f -P
root         787       1  0 14:01 tty1     00:00:00 /sbin/agetty -o -p -- \u --noclear - linux
root         850       1  0 14:02 ?        00:00:00 sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups
root         851     850  0 14:02 ?        00:00:00 sshd: aina [priv]
aina         853     851  0 14:02 ?        00:00:00 sshd: aina@pts/0
aina         854     853  0 14:02 pts/0    00:00:00 -bash
aina         922     854  0 14:03 pts/0    00:00:00 ./mainprocess
aina         923     922  0 14:03 pts/0    00:00:00 ./mainprocess
aina         924     923  0 14:03 pts/0    00:00:00 ./subprocess1
aina         925     922  0 14:03 pts/0    00:00:00 ./mainprocess
aina         926     925  0 14:03 pts/0    00:00:00 ./subprocess2
aina         927     925  0 14:03 pts/0    00:00:00 ./subprocess2
aina         928     925  0 14:03 pts/0    00:00:00 ./subprocess2
aina         929     923  0 14:03 pts/0    00:00:00 ./subprocess1
root         937       2  0 14:07 ?        00:00:00 [kworker/u5:0-events_unbound]
root         941       2  0 14:08 ?        00:00:00 [kworker/1:2]
root         942       2  0 14:08 ?        00:00:00 [kworker/1:3]
aina         943     854  0 14:10 pts/0    00:00:00 ps -ef
```

Question 3

Use the `ps` command with some tools to only list processes named "subprocess" and show some info about them.

Command
`aina@secr2043:~$ ps aux | grep subprocess`
Output

```
aina         924  0.0  0.1   2680  1536 pts/0    S    14:03   0:00 ./subprocess1
aina         926  0.0  0.1   2680  1536 pts/0    S    14:03   0:00 ./subprocess2
aina         927  0.0  0.1   2680  1536 pts/0    S    14:03   0:00 ./subprocess2
aina         928  0.0  0.1   2680  1536 pts/0    S    14:03   0:00 ./subprocess2
aina         929  0.0  0.1   2680  1536 pts/0    S    14:03   0:00 ./subprocess1
aina         950  0.0  0.1   3528  1792 pts/0    S+   14:13   0:00 grep --color=auto subprocess
```

Question 4

Execute the `ps` command, specifying options that reveal only the following columns:

- Process ID (`pid`)
- Owner of the process (`user`)
- CPU percentage (`pcpu`)
- Memory percentage (`pmem`)
- Command (`cmd`)

Command
`aina@secr2043:~$ ps -eo pid,user,pcpu,pmem,cmd`
Output

```
    PID USER      %CPU %MEM CMD
      1 root       0.1  1.3 /sbin/init
      2 root       0.0  0.0 [kthreadd]
      3 root       0.0  0.0 [pool_workqueue_release]
      4 root       0.0  0.0 [kworker/R-rcu_g]
      5 root       0.0  0.0 [kworker/R-rcu_p]
      6 root       0.0  0.0 [kworker/R-slub_]
      7 root       0.0  0.0 [kworker/R-netns]
      8 root       0.0  0.0 [kworker/0:0-cgroup_destroy]
      9 root       0.1  0.0 [kworker/0:1-events]
     10 root       0.0  0.0 [kworker/0:0H-kblockd]
     11 root       0.0  0.0 [kworker/u4:0-floppy]
     12 root       0.0  0.0 [kworker/R-mm_pe]
     13 root       0.0  0.0 [rcu_tasks_kthread]
     14 root       0.0  0.0 [rcu_tasks_rude_kthread]
     15 root       0.0  0.0 [rcu_tasks_trace_kthread]
     16 root       0.0  0.0 [ksoftirqd/0]
     17 root       0.0  0.0 [rcu_preempt]
     18 root       0.0  0.0 [migration/0]
     19 root       0.0  0.0 [idle_inject/0]
     20 root       0.0  0.0 [cpuhp/0]
     21 root       0.0  0.0 [cpuhp/1]
     22 root       0.0  0.0 [idle_inject/1]
     23 root       0.0  0.0 [migration/1]
     24 root       0.0  0.0 [ksoftirqd/1]
     26 root       0.0  0.0 [kworker/1:0H-events_highpri]
     28 root       0.0  0.0 [kworker/u6:0-events_power_efficient]
     29 root       0.0  0.0 [kdevtmpfs]
     30 root       0.0  0.0 [kworker/R-inet_]
     31 root       0.0  0.0 [kworker/u5:1-flush-8:0]
     32 root       0.0  0.0 [kauditd]
     33 root       0.0  0.0 [khungtaskd]
     34 root       0.0  0.0 [oom_reaper]
     35 root       0.0  0.0 [kworker/R-write]
     37 root       0.0  0.0 [kcompactd0]
     38 root       0.0  0.0 [ksmd]
     39 root       0.4  0.0 [kworker/1:1-events]
     40 root       0.0  0.0 [khugepaged]
     41 root       0.0  0.0 [kworker/R-kinte]
     42 root       0.0  0.0 [kworker/R-kbloc]
     43 root       0.0  0.0 [kworker/R-blkcg]
     44 root       0.0  0.0 [irq/9-acpi]
     45 root       0.0  0.0 [kworker/R-tpm_d]
     46 root       0.0  0.0 [kworker/R-ata_s]
     47 root       0.0  0.0 [kworker/R-md]
     48 root       0.0  0.0 [kworker/R-md_bi]
     49 root       0.0  0.0 [kworker/R-edac-]
     50 root       0.0  0.0 [kworker/R-devfr]
     51 root       0.0  0.0 [watchdogd]
     53 root       0.0  0.0 [kworker/1:1H-kblockd]
     54 root       0.0  0.0 [kswapd0]
     55 root       0.0  0.0 [ecryptfs-kthread]
     56 root       0.0  0.0 [kworker/R-kthro]
     57 root       0.0  0.0 [kworker/R-acpi_]
     58 root       0.0  0.0 [kworker/u6:2-events_power_efficient]
     59 root       0.0  0.0 [scsi_eh_0]
     60 root       0.0  0.0 [kworker/R-scsi_]
     61 root       0.0  0.0 [scsi_eh_1]
```

```
   62 root      0.0  0.0 [kworker/R-scsi_]
   64 root      0.0  0.0 [kworker/u6:4-events_power_efficient]
   65 root      0.0  0.0 [kworker/R-mld]
   66 root      0.0  0.0 [kworker/R-ipv6_]
   73 root      0.0  0.0 [kworker/R-kstrp]
   75 root      0.0  0.0 [kworker/u7:0]
   76 root      0.0  0.0 [kworker/u8:0]
   77 root      0.0  0.0 [kworker/u9:0]
   82 root      0.0  0.0 [kworker/R-crypt]
   83 root      0.0  0.0 [kworker/0:1H-kblockd]
   93 root      0.0  0.0 [kworker/R-charg]
  132 root      0.0  0.0 [kworker/u4:1-ext4-rsv-conversion]
  143 root      0.0  0.0 [kworker/R-mpt_p]
  149 root      0.0  0.0 [kworker/R-mpt/0]
  158 root      0.0  0.0 [scsi_eh_2]
  159 root      0.0  0.0 [kworker/R-scsi_]
  193 root      0.0  0.0 [kworker/R-raid5]
  230 root      0.0  0.0 [jbd2/sda1-8]
  231 root      0.0  0.0 [kworker/R-ext4-]
  298 root      0.0  0.0 [psimon]
  303 root      0.0  1.1 /usr/lib/systemd/systemd-journald
  323 root      0.0  0.0 [kworker/u5:3-events_power_efficient]
  333 root      0.0  0.0 [kworker/R-kmpat]
  340 root      0.0  0.0 [kworker/R-kmpat]
  364 root      0.0  2.7 /sbin/multipathd -d -s
  372 root      0.0  0.7 /usr/lib/systemd/systemd-udevd
  384 root      0.0  0.0 [psimon]
  459 root      0.0  0.0 [jbd2/sda16-8]
  460 root      0.0  0.0 [kworker/R-ext4-]
  491 systemd+  0.0  0.9 /usr/lib/systemd/systemd-networkd
  498 systemd+  0.0  1.2 /usr/lib/systemd/systemd-resolved
  500 root      0.0  0.0 [irq/18-vmwgfx]
  501 root      0.0  0.0 [kworker/R-ttm]
  513 systemd+  0.0  0.7 /usr/lib/systemd/systemd-timesyncd
  647 message+  0.0  0.5 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog-o
  651 polkitd   0.0  0.7 /usr/lib/polkit-1/polkitd --no-debug
  657 root      0.0  0.8 /usr/lib/systemd/systemd-logind
  658 root      0.0  1.3 /usr/libexec/udisks2/udisksd
  695 root      0.0  2.3 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
  697 syslog    0.0  0.6 /usr/sbin/rsyslogd -n -iNONE
  712 root      0.0  1.2 /usr/sbin/ModemManager
  779 root      0.0  0.2 /usr/sbin/cron -f -P
  787 root      0.0  0.1 /sbin/agetty -o -p -- \u --noclear - linux
  850 root      0.0  0.8 sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups
  851 root      0.0  0.5 sshd: aina [priv]
  853 aina      0.0  0.7 sshd: aina@pts/0
  854 aina      0.0  0.5 -bash
  922 aina      0.0  0.1 ./mainprocess
  923 aina      0.0  0.0 ./mainprocess
  924 aina      0.0  0.1 ./subprocess1
  925 aina      0.0  0.0 ./mainprocess
  926 aina      0.0  0.1 ./subprocess2
  927 aina      0.0  0.1 ./subprocess2
  928 aina      0.0  0.1 ./subprocess2
  929 aina      0.0  0.1 ./subprocess1
  937 root      0.0  0.0 [kworker/u5:0-events_unbound]
  942 root      0.0  0.0 [kworker/1:3]
  946 root      0.0  0.0 [kworker/0:2-cgroup_destroy]
  951 aina      0.0  0.4 ps -eo pid,user,pcpu,pmem,cmd
```

Question 5

Building on the ps command used in Question 4, can you add an option to sort the listed processes by their memory usage (`pmem`)?

Command
`aina@secr2043:~$ ps -eo pid,user,pcpu,pmem,cmd --sort=-pmem`
Output

```
    PID USER     %CPU %MEM CMD
    364 root      0.0  2.7 /sbin/multipathd -d -s
    695 root      0.0  2.3 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
    658 root      0.0  1.3 /usr/libexec/udisks2/udisksd
      1 root      0.1  1.3 /sbin/init
    498 systemd+  0.0  1.2 /usr/lib/systemd/systemd-resolved
    712 root      0.0  1.2 /usr/sbin/ModemManager
    303 root      0.0  1.1 /usr/lib/systemd/systemd-journald
    491 systemd+  0.0  0.9 /usr/lib/systemd/systemd-networkd
    657 root      0.0  0.8 /usr/lib/systemd/systemd-logind
    850 root      0.0  0.8 sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups
    651 polkitd   0.0  0.7 /usr/lib/polkit-1/polkitd --no-debug
    372 root      0.0  0.7 /usr/lib/systemd/systemd-udevd
    513 systemd+  0.0  0.7 /usr/lib/systemd/systemd-timesyncd
    853 aina      0.0  0.7 sshd: aina@pts/0
    697 syslog    0.0  0.6 /usr/sbin/rsyslogd -n -iNONE
    647 message+  0.0  0.5 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog-o
    851 root      0.0  0.5 sshd: aina [priv]
    854 aina      0.0  0.5 -bash
   1083 aina      0.0  0.4 ps -eo pid,user,pcpu,pmem,cmd --sort=-pmem
    779 root      0.0  0.2 /usr/sbin/cron -f -P
    787 root      0.0  0.1 /sbin/agetty -o -p -- \u --noclear - linux
    922 aina      0.0  0.1 ./mainprocess
    924 aina      0.0  0.1 ./subprocess1
    926 aina      0.0  0.1 ./subprocess2
    927 aina      0.0  0.1 ./subprocess2
    928 aina      0.0  0.1 ./subprocess2
    929 aina      0.0  0.1 ./subprocess1
    923 aina      0.0  0.0 ./mainprocess
    925 aina      0.0  0.0 ./mainprocess
      2 root      0.0  0.0 [kthreadd]
      3 root      0.0  0.0 [pool_workqueue_release]
      4 root      0.0  0.0 [kworker/R-rcu_g]
      5 root      0.0  0.0 [kworker/R-rcu_p]
      6 root      0.0  0.0 [kworker/R-slub_]
      7 root      0.0  0.0 [kworker/R-netns]
      9 root      0.1  0.0 [kworker/0:1-mm_percpu_wq]
     10 root      0.0  0.0 [kworker/0:0H-kblockd]
     11 root      0.0  0.0 [kworker/u4:0-floppy]
     12 root      0.0  0.0 [kworker/R-mm_pe]
     13 root      0.0  0.0 [rcu_tasks_kthread]
     14 root      0.0  0.0 [rcu_tasks_rude_kthread]
     15 root      0.0  0.0 [rcu_tasks_trace_kthread]
     16 root      0.0  0.0 [ksoftirqd/0]
     17 root      0.0  0.0 [rcu_preempt]
     18 root      0.0  0.0 [migration/0]
     19 root      0.0  0.0 [idle_inject/0]
     20 root      0.0  0.0 [cpuhp/0]
     21 root      0.0  0.0 [cpuhp/1]
     22 root      0.0  0.0 [idle_inject/1]
     23 root      0.0  0.0 [migration/1]
     24 root      0.0  0.0 [ksoftirqd/1]
     26 root      0.0  0.0 [kworker/1:0H-events_highpri]
     28 root      0.0  0.0 [kworker/u6:0-events_power_efficient]
     29 root      0.0  0.0 [kdevtmpfs]
     30 root      0.0  0.0 [kworker/R-inet_]
     31 root      0.0  0.0 [kworker/u5:1-events_unbound]
     32 root      0.0  0.0 [kauditd]
     33 root      0.0  0.0 [khungtaskd]
```

```
     34 root       0.0  0.0 [oom_reaper]
     35 root       0.0  0.0 [kworker/R-write]
     37 root       0.0  0.0 [kcompactd0]
     38 root       0.0  0.0 [ksmd]
     39 root       0.4  0.0 [kworker/1:1-mm_percpu_wq]
     40 root       0.0  0.0 [khugepaged]
     41 root       0.0  0.0 [kworker/R-kinte]
     42 root       0.0  0.0 [kworker/R-kbloc]
     43 root       0.0  0.0 [kworker/R-blkcg]
     44 root       0.0  0.0 [irq/9-acpi]
     45 root       0.0  0.0 [kworker/R-tpm_d]
     46 root       0.0  0.0 [kworker/R-ata_s]
     47 root       0.0  0.0 [kworker/R-md]
     48 root       0.0  0.0 [kworker/R-md_bi]
     49 root       0.0  0.0 [kworker/R-edac-]
     50 root       0.0  0.0 [kworker/R-devfr]
     51 root       0.0  0.0 [watchdogd]
     53 root       0.0  0.0 [kworker/1:1H-kblockd]
     54 root       0.0  0.0 [kswapd0]
     55 root       0.0  0.0 [ecryptfs-kthread]
     56 root       0.0  0.0 [kworker/R-kthro]
     57 root       0.0  0.0 [kworker/R-acpi_]
     58 root       0.0  0.0 [kworker/u6:2-events_power_efficient]
     59 root       0.0  0.0 [scsi_eh_0]
     60 root       0.0  0.0 [kworker/R-scsi_]
     61 root       0.0  0.0 [scsi_eh_1]
     62 root       0.0  0.0 [kworker/R-scsi_]
     64 root       0.0  0.0 [kworker/u6:4-flush-8:0]
     65 root       0.0  0.0 [kworker/R-mld]
     66 root       0.0  0.0 [kworker/R-ipv6_]
```

```
     73 root       0.0  0.0 [kworker/R-kstrp]
     75 root       0.0  0.0 [kworker/u7:0]
     76 root       0.0  0.0 [kworker/u8:0]
     77 root       0.0  0.0 [kworker/u9:0]
     82 root       0.0  0.0 [kworker/R-crypt]
     83 root       0.0  0.0 [kworker/0:1H-kblockd]
     93 root       0.0  0.0 [kworker/R-charg]
    132 root       0.0  0.0 [kworker/u4:1-ext4-rsv-conversion]
    143 root       0.0  0.0 [kworker/R-mpt_p]
    149 root       0.0  0.0 [kworker/R-mpt/0]
    158 root       0.0  0.0 [scsi_eh_2]
    159 root       0.0  0.0 [kworker/R-scsi_]
    193 root       0.0  0.0 [kworker/R-raid5]
    230 root       0.0  0.0 [jbd2/sda1-8]
    231 root       0.0  0.0 [kworker/R-ext4-]
    298 root       0.0  0.0 [psimon]
    323 root       0.0  0.0 [kworker/u5:3-events_power_efficient]
    333 root       0.0  0.0 [kworker/R-kmpat]
    340 root       0.0  0.0 [kworker/R-kmpat]
    384 root       0.0  0.0 [psimon]
    459 root       0.0  0.0 [jbd2/sda16-8]
    460 root       0.0  0.0 [kworker/R-ext4-]
    500 root       0.0  0.0 [irq/18-vmwgfx]
    501 root       0.0  0.0 [kworker/R-ttm]
    937 root       0.0  0.0 [kworker/u5:0-events_unbound]
    942 root       0.0  0.0 [kworker/1:3-cgroup_destroy]
    946 root       0.0  0.0 [kworker/0:2-cgroup_destroy]
   1078 root       0.0  0.0 [kworker/1:0]
   1082 root       0.0  0.0 [kworker/u6:1-events_unbound]
```

Question 6

Construct a command using `ps`, suitable options, and any additional tools to visualize the hierarchical structure (tree-like) of the following processes:

- "mainprocess"
- "subprocess1"
- "subprocess2"

Command
`aina@secr2043:~$ ps -o pid,ppid,cmd --forest | grep -E 'mainprocess|subprocess'`

Output

```
    922     854  \_ ./mainprocess
    923     922  |   \_ ./mainprocess
    924     923  |   |   \_ ./subprocess1
    929     923  |   |   \_ ./subprocess1
    925     922  |   \_ ./mainprocess
    926     925  |       \_ ./subprocess2
    927     925  |       \_ ./subprocess2
    928     925  |       \_ ./subprocess2
   1091     854  \_ grep --color=auto -E mainprocess|subprocess
```

Question 7

Use `pstree` command with option that show the number of threads to each process.

Command

```
aina@secr2043:~$ pstree -pT
```

Output

```
systemd(1)-+-ModemManager(712)
           |-agetty(787)
           |-cron(779)
           |-dbus-daemon(647)
           |-multipathd(364)
           |-polkitd(651)
           |-rsyslogd(697)
           |-sshd(850)---sshd(851)---sshd(853)---bash(854)-+-mainprocess(922)-+-mainprocess(923)-+-subprocess1(924)
           |                                               |                  |                  `-subprocess1(929)
           |                                               |                  `-mainprocess(925)-+-subprocess2(926)
           |                                               |                                     |-subprocess2(927)
           |                                               |                                     `-subprocess2(928)
           |                                               `-pstree(1092)
           |-systemd-journal(303)
           |-systemd-logind(657)
           |-systemd-network(491)
           |-systemd-resolve(498)
           |-systemd-timesyn(513)
           |-systemd-udevd(372)
           |-udisksd(658)
           `-unattended-upgr(695)
```

Question 8

Use `renice` command to change priority level of one of process “subprocess1”.

Command

```
aina@secr2043:~$ pgrep -f subprocess1
924
929
aina@secr2043:~$ sudo renice +5 924 929
```

Output

```
924 (process ID) old priority 0, new priority 5
929 (process ID) old priority 0, new priority 5
```

Question 9

Terminate all running processes with the name “mainprocess”.

Command
`aina@secr2043:~$ pkill mainprocess`
Output

`Main process (ID: 925) received signal: 15. Terminating...`
`Main process (ID: 923) received signal: 15. Terminating...`
`aina@secr2043:~$ Main process (ID: 922) received signal: 15. Terminating...`

Question 10

Write a short C or Python code (choose only one language) demonstrating multiprocessing with `fork()` and `wait()`. Compile and/or run the code. Show the output.

Source Code:

```
GNU nano 7.2                                    fork_example.c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    pid_t pid = fork();

    if (pid == 0) {
        printf("Child Process: PID = %d\n", getpid());
    } else {
        wait(NULL);
        printf("Parent Process: PID = %d\n", getpid());
    }

    return 0;
}
```

Output:

```
Child Process: PID = 1127
Parent Process: PID = 1126
```